

KPOW Article



Duncanamps

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This is an email I received from **Jaime Arbona**, who describes a convenient means of placing a power load line on a swept DC simulation for a BJT, Mosfet, or Vacuum Tube. You've seen them before, the curved dotted line which appears on the I_a/V_a chart...

I thought it was a nice tip, so I've placed it here for all to benefit from.

Have you ever needed the maximum power curve on a valve (transistor) device obtained from pspice? What you usually do is to draw it by hand using $I_p = \text{power}/V_p$. Well I got tired of that, so I devised a neat little constant power device (which I call KPOW). You feed it a voltage, it produces a current so that power is constant (and viceversa). I include the subcircuit below:

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* CONSTANT POWER SUBCIRCUIT

.SUBCKT KPOW 1 2

+PARAMS: POWER= 15 ; or any other default

G1 1 2 VALUE={POWER/(0.00001+V(1,2))}

.ENDS KPOW

*\$

The .00001 term is to prevent infinite current when $V(1,2)=0$. Pspice has to be kept happy!

I then made a two terminal graphic device (just a box) using the Symbol Editor and the following parameters:

TEMPLATE=X^@REFDES %1 %2 @model params: power=@power

PART=KPOW

REFDES=KP?

MODEL=KPOW

POWER=15W

When drawing the output characteristics of a valve, just place KPOW in parallel with the (swept) plate source. Click KPOW and enter the max power dissipation of the valve. Then using PROBE, display the current thru KPOW, in addition to the plate current and Presto! you then have a nice constant (max or whatever you like) power curve on your plate characteristics.